

COURSE CODE/TITLE:	CPE 212- Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/19/2026	DATE SUBMITTED:	09/19/2026
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ACTIVITY NO. 8.1 Install, Configure, and Manage Enterprise Availability Monitoring via Ansible

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

```
(base) jerald@workstation:~$ cd ~
(base) jerald@workstation:~$ mkdir -p nagios-ansible-role
(base) jerald@workstation:~$ cd nagios-ansible-role
(base) jerald@workstation:~/nagios-ansible-role$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/jerald/nagios-ansible-role/.git/
```

Create the dedicated project directory nagios-ansible-role on the control node workstation, navigate into it, and initialize a new Git repository. Set the default main branch to main to establish version control tracking.

```
(base) jerald@workstation:~/nagios-ansible-role$ git branch -M main
(base) jerald@workstation:~/nagios-ansible-role$ mkdir -p roles
(base) jerald@workstation:~/nagios-ansible-role$ ansible-galaxy role init roles/nagios
- Role roles/nagios was created successfully
(base) jerald@workstation:~/nagios-ansible-role$ nano inventory.yaml
```

Create the roles directory and use ansible-galaxy to automatically generate a standardized Ansible role framework named nagios.

```
GNU nano 7.2    inventory.yaml
all:
  children:
    ubuntu_hosts:
      hosts:
        192.168.56.101:
          ansible_user: jeraldinterno
        192.168.56.103:
          ansible_user: interno
    centos_hosts:
      hosts:
        192.168.56.104:
          ansible_user: internos2
    nagios_servers:
      children:
        ubuntu_hosts:
        centos_hosts:
```

Create the inventory.yaml file in the root directory. Group target hosts into ubuntu_hosts (.101, .103) and centos_hosts (.104), and combine them under the parent group nagios_servers with their respective SSH access credentials.

```
(base) jerald@workstation:~/nagios-ansible-role$ nano site.yaml
```

```
GNU nano 7.2    site.yml
--
- name: Deploy Nagios Core on Ubuntu and CentOS
  hosts: nagios_servers
  become: true
  roles:
    - nagios
```

Create the main execution entry point site.yml targeting the nagios_servers group, enabling elevated privileges (become: true), and calling the custom nagios role.

```
(base) jerald@workstation:~/nagios-ansible-role$ nano roles/nagios/tasks/main.yml
```

```
GNU nano 7.2                                roles/nagios/tasks/main.yml
- name: Install dependencies for Ubuntu/Debian
  ansible.builtin.apt:
    name:
      - build-essential
      - apache2
      - php
      - libapache2-mod-php
      - libgd-dev
      - libssl-dev
      - unzip
      - tar
      - make
      - gcc
    state: present
    update_cache: yes
    when: ansible_facts['os_family'] == "Debian"

- name: Install dependencies for CentOS/RedHat
  ansible.builtin.package:
    name:
      - gcc
      - glibc
      - glibc-common
      - make
      - gettext
      - automake
      - autoconf
      - wget
```

Edit roles/nagios/tasks/main.yml to define the installation procedure. Leverage `ansible_facts['os_family']` to conditionally install OS-specific dependencies (apt for Debian/Ubuntu, package for RedHat/CentOS), create system accounts/groups (nagios/nagcmd), download and compile Nagios Core source code with OpenSSL headers, configure web interfaces, and enable background system services.

```
(base) jerald@workstation:~/nagios-ansible-role$ ansible-playbook -i inventory.yaml site.yml -k -K
\SSH password:
BECOME password[defaults to SSH password]:

PLAY [Deploy Nagios Core on Ubuntu and CentOS] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.104]
ok: [192.168.56.101]

TASK [nagios : Install dependencies for Ubuntu/Debian] *****
skipping: [192.168.56.101]
changed: [192.168.56.103]
changed: [192.168.56.104]

TASK [nagios : Install dependencies for CentOS/RedHat] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.101]

TASK [nagios : Create nagios user group] *****
ok: [192.168.56.103]
ok: [192.168.56.104]
ok: [192.168.56.101]

TASK [nagios : Create nagios user] *****
ok: [192.168.56.103]
ok: [192.168.56.104]
ok: [192.168.56.101]

changed: [192.168.56.101] => (item=install-config)
changed: [192.168.56.101] => (item=install-webconf)

TASK [nagios : Enable Apache CGI module (Debian)] *****
skipping: [192.168.56.101]
changed: [192.168.56.104]
changed: [192.168.56.103]

TASK [nagios : Start and Enable Web Server (Debian)] *****
skipping: [192.168.56.101]
ok: [192.168.56.103]
ok: [192.168.56.104]

TASK [nagios : Start and Enable Web Server (RedHat)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.101]

TASK [nagios : Start and Enable Nagios Service] *****
ok: [192.168.56.101]
changed: [192.168.56.103]
changed: [192.168.56.104]

PLAY RECAP *****
192.168.56.101      : ok=12  changed=5  unreachable=0  failed=0  skipped=4  rescued=0
  ignored=0
192.168.56.103      : ok=13  changed=8  unreachable=0  failed=0  skipped=3  rescued=0
  ignored=0
192.168.56.104      : ok=13  changed=8  unreachable=0  failed=0  skipped=3  rescued=0
  ignored=0
```

Execute the playbook across all target nodes using interactive authentication

```
(base) jerald@workstation:~/nagios-ansible-role$ ansible nagios_servers -i inventory.yaml -m ansible.b
uiltin.command -a "systemctl status nagios" -k -K
SSH password:
BECOME password[defaults to SSH password]:
192.168.56.104 | CHANGED | rc=0 >>
● nagios.service - Nagios Core 4.4.14
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-19 15:12:45 UTC; 1min 27s ago
     Docs: https://www.nagios.org/documentation
   Process: 16344 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg (code
=exited, status=0/SUCCESS)
   Process: 16345 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg (code=ex
ited, status=0/SUCCESS)
   Main PID: 16347 (nagios)
    Tasks: 6 (limit: 3465)
   Memory: 5.7M (peak: 6.1M)
      CPU: 342ms
   CGroup: /system.slice/nagios.service
           └─16347 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             └─16348 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
               └─16349 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                 └─16350 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                   └─16351 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                     └─16364 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
192.168.56.103 | CHANGED | rc=0 >>
● nagios.service - Nagios Core 4.4.14
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-19 23:12:43 PST; 1min 30s ago
     Docs: https://www.nagios.org/documentation
```

Verify service execution using an ad-hoc module across all managed nodes

```
(base) jerald@workstation:~/nagios-ansible-role$ git config --global user.name
JeraldSanInterno
(base) jerald@workstation:~/nagios-ansible-role$ git config --global user.email
qjsminterno@tip.edu.ph
(base) jerald@workstation:~/nagios-ansible-role$ git add .
(base) jerald@workstation:~/nagios-ansible-role$ git status
On branch main
Changes to be committed:
  Help: "git restore --staged <file>..." to unstage)
        modified:   roles/nagios/tasks/main.yml

(base) jerald@workstation:~/nagios-ansible-role$ git commit -m "Complete Ansible Nagios role deploymen
t for Ubuntu and CentOS"
[main 80efcf7] Complete Ansible Nagios role deployment for Ubuntu and CentOS
 1 file changed, 6 deletions(-)
(base) jerald@workstation:~/nagios-ansible-role$ git commit -m "Complete Ansible Nagios role deploymen
t for Ubuntu and CentOS"
On branch main
nothing to commit, working tree clean
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

SUPPLEMENTARY ACTIVITY

1. What are the benefits of having an availability monitoring tool?

An availability monitoring tool like Nagios provides real-time visibility into server and service uptime, allowing system administrators to detect network outages and infrastructure degradation immediately. By automatically sending alerts when critical services fail, it drastically reduces Mean Time to Repair (MTTR) and minimizes costly system downtime before end users are impacted. Additionally, it collects historical performance data that aids in capacity planning, resource optimization, and ensuring compliance with Service Level Agreements (SLAs).

Conclusions/Learnings:

Through this activity, I learned how to automate the configuration and deployment of Nagios Core across a multi-distribution Linux environment using Ansible roles. I gained practical experience resolving cross-platform package management differences between Debian and RedHat operating systems, handling source compilation dependencies, and troubleshooting file system permission constraints in automated workflows. Finally, I reinforced best practices in version control by committing and publishing the finalized modular configuration to GitHub.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: idea generation, grammar correction, code assistance, explanation of concepts

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.